

Module 4 Summary – Jacob Booth 221530207

Response to feedback

New upload post-submission but not a resubmission, based on feedback from module 3. I've leaned into developing the overall picture as I introduce new concepts. I got rid of the rephrasing and streamlined the summary as per feedback from the previous resubmission.

Module Learning Objectives

I certify that I achieved the following learning objectives for the module:

- Calculate determinants using the cofactor expansion
- Use properties of determinants (Theorem 3.1.2) to calculate determinants
- Calculate matrix inverses using $\text{adj } A$ and $\det(A)$
- Identify the inverse of an orthogonal matrix
- Perform LU-factorisation on a matrix and use this to solve systems of linear equations

Key Definitions and Theorems

The Cofactor Expansion

Textbook Ref	Theorem/Definition statement	Learning Notes
Determinant Calculation for a 3×3 matrix	For a 3×3 matrix, A – the <u>determinant</u> is given by $\det A = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$ $= aei + bfg + cdh - ceg - afh - bdi$ $= a(ei - fh) - b(di - fg) + c(dh - eg)$ $= a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$	Draws on the products of diagonals Use either the shortcut formula with Sarrus' rule or expansion by minors
Definition 3.1 Cofactors of a matrix	Assume that determinants of $(n - 1) \times (n - 1)$ matrices have been defined. Given the $n \times n$ matrix A, let A_{ij} denote the $(n - 1) \times (n - 1)$ matrix obtained from A by deleting row i and column j. Then the (i, j)-<u>cofactor</u> $c_{ij}(A)$ is the scalar defined by $c_{ij}(A) = (-1)^{i+j} \det(A_{ij})$ Here $(-1)^{i+j}$ is called the <u>sign</u> of the (i, j)-position.	Dissecting sub-matrices from larger ones, carrying the idea of composition Convenient method, just requires reference to the formula

Definition 3.2 Cofactor expansion of a matrix	Assume that determinants of $(n - 1) \times (n - 1)$ matrices have been defined. If $A = [a_{ij}]$ is $n \times n$ define, $\det A = a_{11}c_{11}(A) + a_{12}c_{12}(A) + \dots + a_{1n}c_{1n}(A)$ This is called the <u>cofactor expansion</u> of $\det A$ along row 1.	Defining the determinant from smaller pieces, recursively defining smaller matrix dets to define the det of A
Theorem 3.1.2 Determinant properties	Let A denote an $n \times n$ matrix. 1. If A has a row or column of zeros, $\det A = 0$. 2. If two distinct rows (or columns) of A are interchanged, the determinant of the resulting matrix is $-\det A$. 3. If a row (or column) of A is multiplied by a constant u , the determinant of the resulting matrix is $u(\det A)$. 4. If two distinct rows (or columns) of A are identical, $\det A = 0$. 5. If a multiple of one row of A is added to a different row (or if a multiple of a column is added to a different column), the determinant of the resulting matrix is $\det A$.	Unique behaviour of det, proportional scaling but adding one row to another doesn't change det Reduces the logic down to 5 constraints on the determinant, helpful for quick interpretation

Determinants and matrix inverses

Textbook Ref	Theorem/Definition statement	Notes
Theorem 3.2.1 Product theorem	If A and B are $n \times n$ matrices, then $\det(AB) = \det A \det B$.	Matrix product det or det of matrix product, both are the same
Theorem 3.2.2 Det relationship to inverse	An $n \times n$ matrix A is invertible if and only if $\det A \neq 0$. When it is invertible, we have, $\det(A^{-1}) = \frac{1}{\det A}$	Any matrix with a non-zero det is invertible
Theorem 3.2.3 Det of transpose	If A is any square matrix, $\det(A^T) = \det A$	Det persists across transposition
Theorem 3.2.4 Adjugate formula	If A is any square matrix, then $A(\text{adj } A) = (\det A) I = (\text{adj } A) A$ In particular, if $\det A \neq 0$, the inverse of A is given by $A^{-1} = \frac{1}{\det A} \text{adj } A$	Multiplying A with $\text{adj}(A)$ gives $(\det(A))I$ Leading into the inverse formula

LU Factorisation

Textbook Ref	Theorem/Definition statement	Notes
Theorem 2.7.1 LU Theorem	If A can be lower reduced to a row-echelon matrix U, then $A=LU$ where L is lower triangular and invertible and U is upper triangular and row-echelon.	Without row swaps, if matrix A can be reduced then it factors into L and U composition
LU Algorithm	Let A be an $m \times n$ matrix of rank r, and suppose that A can be lower reduced to a row-echelon matrix U . Then $A = LU$ where the lower triangular, invertible matrix L is constructed as follows: 1. If $A=0$, take $L=I_m$ and $U=0$. 2. If $A \neq 0$, write $A_1=A$ and let $\mathbf{c}_1$ be the leading column of A_1. Use $\mathbf{c}_1$ to create the first leading 1 and create zeros below it (using lower reduction). When this is completed, let A_2 denote the matrix consisting of rows 2 to m of the matrix just created. 3. If $A_2 \neq 0$, let $\mathbf{c}_2$ be the leading column of A_2 and repeat Step 2 on A_2 to create A_3. 4. Continue in this way until U is reached, where all rows below the last leading 1 consist of zeros. This will happen after r steps. 5. Create L by placing $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_r$ at the bottom of the first r columns of I_m.	For each multiplier in elimination, build L from each m, recording scalar pivots in L and reduced A in U

Summarising your understanding:

Module 4 expresses the behaviour of determinants in the scope of cofactors and compositions. The module establishes cofactor expansion as the foundation for determinant calculation, a multi-faceted metric for inferring matrix behaviour in context. Module 4 compares the preservation/scaling properties of each elementary operation, the matrix adjugate and transposition are reintroduced to highlight the behaviour of the determinant's consistency across transformed, inverted, and transposed matrices.

4.1 The Cofactor Expansion

Cofactor expansion recursively builds on sub-matrices, multiplying nested matrices by the sign $(-1)^{i+j}$, defined as the **cofactor** by **Definition 3.1**. Summing the entry-products with corresponding factors returns the determinant. This ties directly to the behaviour of determinants, expressed by the structural properties in **Theorem 3.1.2**. Similarly to how the previous module introduced inverse theorem and inverse properties as the framework for the inversion algorithm, **determinant properties** outline the behaviour of the determinant under elementary row operations, which is key to how **LU Factorisation** approaches reduction (no row swaps).

Include a worked example of applying some of the rules in Theorem 3.1.2 to simplify the calculation of a larger determinant.

The example below leverages **determinant properties** to perform quick mental-shortcuts for calculating the determinant. Based on condition 2, swapping two rows or columns brings $\det(A)$ to $-\det(A)$, immediately clarifying the result if $\det(A)$ is known. Condition 5 is leveraged in the second problem for clarifying that there is no change in $\det(A)$ after reduction, adding a multiple of one row/column to another doesn't alter the determinant.

N4.2.2

When $\det(A) = 9$, find $\det(B)$

Swap 1 $\leftrightarrow$ 4, 2, 3 unchanged

$$\therefore -\det(A) \Rightarrow \det(B) = -\det(A) = -9$$

Using Theorem 3.1.2

Condition 2, $-\det(A)$

N4.2.3

$A = \begin{pmatrix} 2 & 0 & -3 & 2 \\ 4 & 2 & 5 & -2 \\ 0 & 2 & -1 & -3 \\ 6 & 0 & 4 & 5 \\ 0 & 0 & 2 & -1 \\ 2 & 6 & 2 & -14 \\ 0 & 6 & 1 & 12 \end{pmatrix}$
 Step 1: $R_1 \leftrightarrow R_6$, $R_2 \leftrightarrow R_7$, $R_3 \leftrightarrow R_5$

$B = \begin{pmatrix} 2 & 0 & -3 & 2 \\ 0 & -2 & -1 & -3 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & 0 & -4 \end{pmatrix}$
 Step 2: $R_1 \leftrightarrow R_4$, $R_2 \leftrightarrow R_3$
 Step 3: $R_1 \leftrightarrow R_4$

Property 5, $\det$ unaltered

$$\therefore \det(A) = (-2)(-2)(-2)(-4) = 32$$

Summarily, **determinant properties** are leveraged for guiding the calculation, and when necessary, locally-scoped sub-matrices can be used to infer linear dependence at the global-level.

4.2 Determinants and Matrix Inverses

Building from **cofactor expansion**, the determinant is expressed as a sign for whether the matrix can be inverted, linking directly back to the previous module. **Theorem 3.2.2** states that a matrix can only be inverted if the determinant is nonzero. **Theorems 3.2.1, 3.2.3, and 3.2.4** demonstrate the behaviour of the determinant when exposed to multiplication, transposition, and the adjugate, providing tools for calculating the determinant in the problem domain, consistent with **determinant properties**.

Summarise Cramer's rule and how it works. Include reference to any of the relevant definitions and theorems.

Cramer's Rule is a method for solving a linear system given the determinant is nonzero (**Theorem 3.2.2**), and the coefficient matrix is square. For each x_i variable in the system, replace the column i with the constant vector b , giving an intermediate matrix A_i . Each column of the coeff matrix is iteratively replaced with the constant vector, and each locally-calculated determinant gets divided by the original coeff determinant, returning the unknown values. Again, **Theorems 3.2.1, 3.2.3, and 3.2.4** express the behaviour of determinants in context, and how the inverse, transpose, and adjugate affect it. Cramer's Rule leverages their natural behaviour for methodically solving the unknown variables.

Tying these ideas together, **Theorem 3.2.3** clarifies that transposition has no alteration on the determinant, and **Theorem 3.1.2** treats row/col perspectives the same. This symmetry shows that cofactor expansion along either a row or column is valid, verified by the adjugate matrix dependence on transposing the cofactor matrix. Cramer's rule works under this behaviour and the other unique **determinant properties** to solve linear systems.

4.3 LU Factorisation

Building from the modular, recursive approach of **cofactor expansion**, and the behaviour of **determinant properties**, LU factorisation ties these concepts together to transform the problem $Ax = b$ into $LUx = b$. This composes the solution into two sub-matrices, whose factor is the single-matrix solution. Drawing directly from **Theorem 3.1.2 (5)**, similarly to the previous worked example, avoiding row swaps leaves the determinant unaltered, so **LU Theorem** is consistent with **determinant properties**. The **LU Algorithm** works under the conditional framework established by determinant properties, using Gaussian Elimination to form a composition of a scalar matrix L , and reduced matrix U .

Write a statement on LU factorisation and when it can be applied

LU factorisation expresses the problem $Ax = b$ as $LUx = b$. It's an algorithm that can be applied to full-rank square matrices, so long as they can be reduced to an upper-triangular without any row swaps. LU Factorisation decomposes a square matrix A into the product of

a lower-triangular matrix L and upper-triangular matrix U , expressing A as a factor of LU . This simplifies the problem space for solving $Ax = b$ by reducing the matrix A into two smaller triangular systems.

Similarly to the last, module 4 encapsulates the behaviour of focus-specific properties to paint the larger picture of how an algorithm works. LU factorisation operates under the behaviour established by **determinant properties** in the context of inversion, products, the adjugate, and transposition, to aggregate the solution from a composition of sub-problems for full-rank matrices. Summarily, each property respects the behaviour of other properties, and establishes unique behaviour that's consistent with each other. LU factorisation leverages the interdependence to maintain solution consistency, tying back to the idea of large contextual problems composed of smaller locally-scoped tasks.

Reflecting on the content

I found this module more intuitive than it initially appeared. I received a hit of nostalgia performing cofactor expansion and 3×3 determinant calculation, methods I'd seen and used in past classes without establishing the link until I was solving the problems. This was helpful and encouraging. I can also relate to the necessity of these concepts as a problem-solver from a computer science background. Breaking down large problems into meaningful, functional components forms the backbone of robust software engineering, in addition to the key module concepts being directly applicable to linear data analysis and software speedup. Overall, a very enjoyable unit that simplifies all the granular processes we've been working with weeks 1 – 4.

References

Summary, notes

[1] J. Chasnov, "LU Decomposition," Applied Linear Algebra and Differential Equations, LibreTexts, [Online].

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Verification

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